Linear Algebra – Problem Set 6 — Solutions

Eigenvalues & Eigenvectors · Full solutions · Original problems in the spirit of MIT 18.06 (Strang)

Two habits run through every solution below. First, **eigenvectors are verified, never derived in silence**: once a candidate is produced, one multiplication $A\mathbf{v}$ settles whether it is right, and that check is written out each time. Second, **trace and determinant are used as a standing audit** — the eigenvalues must sum to $\operatorname{tr} A$ and multiply to $\det A$ — which is why several problems ask for the spectrum without ever expanding $\det(A-\lambda I)$. Where a problem exposes a misconception (Problems 4, 7, 11) the counterexample is the answer; keep the specific numbers ($(2,-1)$, the shear $\begin{bmatrix}2&1\\0&2\end{bmatrix}$, the rotation $\begin{bmatrix}0&-1\\1&0\end{bmatrix}$) in memory, since they settle almost every tempting false statement.

Problem 1

$A=\begin{bmatrix}3&0\\0&-1\end{bmatrix}$. (a) Test four vectors. (b) Why $\mathbf{v}\neq\mathbf{0}$ in the definition. (c) Geometric picture.

(a) Multiply each one out:

$$A\begin{bmatrix}1\\0\end{bmatrix}=\begin{bmatrix}3\\0\end{bmatrix}=3\begin{bmatrix}1\\0\end{bmatrix},\qquad A\begin{bmatrix}0\\1\end{bmatrix}=\begin{bmatrix}0\\-1\end{bmatrix}=(-1)\begin{bmatrix}0\\1\end{bmatrix},\qquad A\begin{bmatrix}1\\1\end{bmatrix}=\begin{bmatrix}3\\-1\end{bmatrix},\qquad A\begin{bmatrix}2\\0\end{bmatrix}=\begin{bmatrix}6\\0\end{bmatrix}=3\begin{bmatrix}2\\0\end{bmatrix}.$$

So $(1,0)$ and $(2,0)$ are eigenvectors with $\lambda=3$ (they lie on the same line, which is why the second one adds nothing new), $(0,1)$ is an eigenvector with $\lambda=-1$, and $(1,1)$ is *not* an eigenvector: $(3,-1)=c(1,1)$ would force $c=3$ and $c=-1$ at once. Eigenvalues found: $\lambda=3$ (direction $(1,0)$) and $\lambda=-1$ (direction $(0,1)$). **(b)** The definition requires the eigenvector to be **non-zero**. If $\mathbf{0}$ were allowed, then $A\mathbf{0}=\mathbf{0}=\lambda\mathbf{0}$ would hold for every scalar λ, so every matrix would have every number as an eigenvalue. The spectrum would carry zero information, and every theorem that depends on it would collapse: there would be no finite list of n roots to speak of, "$\det A=$ product of the eigenvalues" and "$\operatorname{tr} A=$ sum of the eigenvalues" would be meaningless, and "there is a basis of eigenvectors" would be trivially true everywhere. Insisting on $\mathbf{v}\neq\mathbf{0}$ is exactly what makes the eigenvalue list finite and worth computing. **(c)** Corners go $(0,0)\mapsto(0,0)$, $(1,0)\mapsto(3,0)$, $(1,1)\mapsto(3,-1)$, $(0,1)\mapsto(0,-1)$: the unit square becomes a 3×1 rectangle below the x-axis. Horizontally everything is stretched by 3; vertically everything is flipped through the origin (length unchanged, sign reversed). The horizontal direction therefore stays on itself, and the vertical direction also stays on its *line* — being reversed still means $A\mathbf{v}$ lies on the same line through the origin, which is all $A\mathbf{v}=\lambda\mathbf{v}$ asks for, with $\lambda=-1$. Every other direction fails: the line $y=x$ lands along $(3,-1)$, slope $-\frac{1}{3}\neq1$.

Problem 2

Eigenvalues without the characteristic polynomial, for $B=\begin{bmatrix}2&1\\1&2\end{bmatrix}$ and $C=\begin{bmatrix}0&1&1\\1&0&1\\1&1&0\end{bmatrix}$.

(a) Just multiply the two obvious symmetric directions:

$$B\begin{bmatrix}1\\1\end{bmatrix}=\begin{bmatrix}3\\3\end{bmatrix}=3\begin{bmatrix}1\\1\end{bmatrix},\qquad B\begin{bmatrix}1\\-1\end{bmatrix}=\begin{bmatrix}1\\-1\end{bmatrix}=1\cdot\begin{bmatrix}1\\-1\end{bmatrix}.$$

So $\lambda=3$ (direction $(1,1)$) and $\lambda=1$ (direction $(1,-1)$) — no determinant expansion needed. **(b)** Two facts close the door on anything else. (i) $\det(B-\lambda I)$ is a polynomial of degree 2 in λ,

so counting multiplicity there are exactly 2 eigenvalues and no more; we have already found two *distinct* ones. (ii) The two identities act as an audit: $\lambda_1 + \lambda_2 = 3 + 1 = 4 = \operatorname{tr} B = 2 + 2$ and $\lambda_1\lambda_2 = 3 \cdot 1 = 3 = \det B = 4 - 1$, both matching. So the list $\{1, 3\}$ is complete. **(c)** If every row of A sums to s, then the i-th entry of $A\mathbf{1}$ is (row i of A) $\cdot\, \mathbf{1} = s$, i.e.

$$A \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} = \begin{bmatrix} s \\ \vdots \\ s \end{bmatrix} = s \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix},$$

so $(1, \dots, 1)$ is an eigenvector with eigenvalue s — the vector of all ones always "sums up" a row. Here: $C(1,1,1) = (1+1,\ 1+1,\ 1+1) = (2,2,2) = 2(1,1,1)$, so $\lambda = 2$. (Note the columns also sum to 2, which is not what we used — the statement is about rows.) **(d)** For a 3×3 matrix $-\lambda^3 + (\operatorname{tr} C)\lambda^2 - \cdots$ shows $\lambda_1 + \lambda_2 + \lambda_3 = \operatorname{tr} C = 0$ and $\lambda_1\lambda_2\lambda_3 = \det C = 2$. With $\lambda_1 = 2$: the other two satisfy $\lambda_2 + \lambda_3 = -2$ and $\lambda_2\lambda_3 = 2/2 = 1$, so they are the roots of $x^2 + 2x + 1 = (x+1)^2$. Hence the remaining eigenvalues coincide: Spectrum of C: $\{2, -1, -1\}$ (with -1 of algebraic multiplicity 2). Audit: $2 + (-1) + (-1) = 0 = \operatorname{tr} C$ and $2(-1)(-1) = 2 = \det C$. The moral of (a)–(c): verifying a guessed eigenvector costs one multiplication, whereas grinding the polynomial costs half a page. Structure (symmetry, equal row sums, diagonal shape) is usually telling you the answer for free.

Problem 3 — Full procedure for $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$

(a) characteristic polynomial, (b) eigenvalues with the trace/determinant audit, (c) both eigenspaces, (d) verification, (e) what rank 0 would have meant.

(a) Subtracting λ touches only the diagonal:

$$A - \lambda I = \begin{bmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{bmatrix}, \qquad \det(A - \lambda I) = (4-\lambda)(3-\lambda) - 2\cdot 1 = \lambda^2 - 7\lambda + 12 - 2 = \lambda^2 - 7\lambda + 10 = (\lambda - 2)(\lambda - 5).$$

(b) Characteristic equation $(\lambda-2)(\lambda-5) = 0$ gives $\lambda = 2$ and $\lambda = 5$. Audit: sum $= 7 = \operatorname{tr} A = 4+3$ ✓, product $= 10 = \det A = 12 - 2$ ✓. **(c)** Each eigenvalue gets its own substitution and its own reduction. $\lambda = 5$: $A - 5I = \begin{bmatrix} -1 & 1 \\ 2 & -2 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & -1 \\ 0 & 0 \end{bmatrix}$, so $-x + y = 0$, i.e. $y = x$ with x free:

$$\mathbf{v} = t \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad t \neq 0.$$

$\lambda = 2$: $A - 2I = \begin{bmatrix} 2 & 1 \\ 2 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 2 & 1 \\ 0 & 0 \end{bmatrix}$, so $2x + y = 0$, i.e. $y = -2x$ with x free:

$$\mathbf{v} = t \begin{bmatrix} 1 \\ -2 \end{bmatrix}, \quad t \neq 0.$$

Eigenspace $E_5 = \operatorname{Span}\{(1,1)\}$, $\quad E_2 = \operatorname{Span}\{(1,-2)\}$. Each is a *line*: write answers with the free parameter and remember $t \neq 0$ (Problem 1(b)). **(d)** Two multiplications:

$$A \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 4+1 \\ 2+3 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix} = 5 \begin{bmatrix} 1 \\ 1 \end{bmatrix} \ \checkmark, \qquad A \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 4-2 \\ 2-6 \end{bmatrix} = \begin{bmatrix} 2 \\ -4 \end{bmatrix} = 2 \begin{bmatrix} 1 \\ -2 \end{bmatrix} \ \checkmark.$$

This step is not optional because it is the *definition* — everything before it was bookkeeping. It catches exactly the errors that are easy to make and hard to see: a sign slip in expanding

$(4-\lambda)(3-\lambda)$, a wrong constant term (it is -2, from the off-diagonal product), and the classic mix-up of pairing $\lambda = 5$ with the vector computed for $\lambda = 2$. **(e)** In $\mathbb{R}^2$ an eigenspace of dimension 1 needs $\operatorname{rank}(A - \lambda I) = 1$: one pivot, one free variable, a line of eigenvectors. If $A - \lambda I$ had come out rank 0 it would be the zero matrix, meaning $A = \lambda I$ itself — then every non-zero vector is an eigenvector, the eigenspace is the whole plane, and this λ is the only eigenvalue. (Rank 2 would be the other failure: only $\mathbf{v} = \mathbf{0}$ solves it, so "λ" was never an eigenvalue.)

Problem 4

Eigenspaces versus the set of eigenvectors.

(a) Directly, using nothing but linearity of A. Take $\mathbf{v}, \mathbf{w} \in E_\lambda$ and scalars c, d:

$$A(c\mathbf{v} + d\mathbf{w}) = c\,A\mathbf{v} + d\,A\mathbf{w} = c(\lambda\mathbf{v}) + d(\lambda\mathbf{w}) = \lambda(c\mathbf{v} + d\mathbf{w}).$$

So $c\mathbf{v} + d\mathbf{w}$ is again in E_λ (and $A\mathbf{0} = \mathbf{0} = \lambda\mathbf{0}$ puts $\mathbf{0}$ in E_λ, which is why the definition of E_λ adds it explicitly). Closure under addition and scalar multiplication: E_λ is a subspace. **(b)** Rewrite the defining equation:

$$A\mathbf{v} = \lambda\mathbf{v} \iff A\mathbf{v} - \lambda\mathbf{v} = \mathbf{0} \iff (A - \lambda I)\mathbf{v} = \mathbf{0} \iff \mathbf{v} \in \operatorname{Nul}(A - \lambda I).$$

So E_λ *is* a null space, and null spaces are subspaces automatically — part (a) was guaranteed all along. This identification is also the practical route to computing eigenvectors: find the special solutions of $(A - \lambda I)\mathbf{v} = \mathbf{0}$, exactly as in Problem 3(c). **(c)** The sum is $(1,1) + (1,-2) = (2,-1)$, and

$$A\begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 8-1 \\ 4-3 \end{bmatrix} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}.$$

For $(7,1)$ to be a multiple of $(2,-1)$ we would need $7 = 2c$ and $1 = -c$, i.e. $c = 7/2$ and $c = -1$ simultaneously — impossible. So the sum of two eigenvectors need not be an eigenvector, and consequently $E_5 \cup E_2$ is not a subspace: it contains $(1,1)$ and $(1,-2)$ but not their sum. (Each piece separately *is* a subspace; taking the union destroys closure.) **(d)** The correct statement: *eigenvectors belonging to the same eigenvalue form a subspace, and only then can you add them freely.* If $\mathbf{v} \in E_\lambda$ and $\mathbf{w} \in E_\lambda$, part (a) gives $\mathbf{v}+\mathbf{w} \in E_\lambda$. If they belong to *different* eigenvalues, the sum generally is not an eigenvector at all — as (c) shows, and once independence of eigenvectors from distinct eigenvalues is available (Problem 6) one can show it never is: from $A(\mathbf{v}+\mathbf{w}) = \nu(\mathbf{v}+\mathbf{w})$ we would get $(\lambda - \nu)\mathbf{v} + (\mu - \nu)\mathbf{w} = \mathbf{0}$, forcing $\lambda = \mu = \nu$.

Problem 5

$\lambda = 0$, singularity, and reading eigenvalues off trace and determinant.

(a) Chain the equivalences one at a time:

0 is an eigenvalue $\iff \exists\,\mathbf{v} \neq \mathbf{0},\ A\mathbf{v} = 0\mathbf{v} = \mathbf{0} \iff$ Nul(A) contains a non-zero vector $\iff A$ is not invertible $\Leftarrow$

The middle link is the invertible-matrix theorem (for a square matrix: trivial null space $\iff$ invertible), and it is precisely what "singular" means. **(b)** Row 2 is twice row 1, so $\det M = 0$ and by (a) one eigenvalue is 0. Then $\operatorname{tr} M = 1 + 4 = 5 = \lambda_1 + \lambda_2 = 0 + \lambda_2$, so the other is 5. $\lambda = 0$: solve $M\mathbf{v} = \mathbf{0}$, i.e. $x + 2y = 0 \to \mathbf{v} = t(-2,1)$. Check: $M(-2,1) = (-2+2, -4+4) = (0,0)$ ✓. $\lambda = 5$: $M - 5I = \begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix} \to \begin{bmatrix} -4 & 2 \\ 0 & 0 \end{bmatrix}$, so $-4x + 2y = 0$, i.e. $y = 2x \to \mathbf{v} = t(1,2)$. Check:

$M(1,2) = (1+4,\ 2+8) = (5,10) = 5(1,2)$ ✓. Eigenvalues $\{0, 5\}$; eigenspaces Span$\{(-2,1)\}$ and Span$\{(1,2)\}$. **(c)** From the trace: $\lambda_1 + \lambda_2 + \lambda_3 = \operatorname{tr} N = 6$, so the third eigenvalue is $6-1-2=3$. Consistency check with the determinant: $1 \cdot 2 \cdot 3 = 6 = \det N$ ✓. No eigenvalue is 0 (equivalently $\det N = 6 \neq 0$), so Third eigenvalue 3, and N is invertible. Eigenvalues of N^{-1}: 1, $\frac{1}{2}$, $\frac{1}{3}$ (reciprocals, since $N\mathbf{v} = \lambda\mathbf{v} \Rightarrow N^{-1}\mathbf{v} = \lambda^{-1}\mathbf{v}$). Eigenvalues of N^2: 1, 4, 9 (squares, since $N^2\mathbf{v} = \lambda^2\mathbf{v}$). **(d)** Because $\det A = \lambda_1\lambda_2\cdots\lambda_n$, a product vanishes exactly when one of its factors does: $\det A = 0 \iff$ some $\lambda_i = 0$. That restatement of part (a) is why computing a single determinant already answers the invertibility question, long before any eigenvector appears.

Problem 6

Distinct eigenvalues force independence of eigenvectors.

(a) Suppose $c\mathbf{v} + d\mathbf{w} = \mathbf{0}$. Apply A to both sides:

$$c\,A\mathbf{v} + d\,A\mathbf{w} = \mathbf{0} \quad \Longrightarrow \quad c\lambda\mathbf{v} + d\mu\mathbf{w} = \mathbf{0}.$$

Now multiply the original equation by λ and subtract it:

$$(c\lambda\mathbf{v} + d\mu\mathbf{w}) - (c\lambda\mathbf{v} + d\lambda\mathbf{w}) = \mathbf{0} \quad \Longrightarrow \quad d(\mu-\lambda)\mathbf{w} = \mathbf{0}.$$

Since $\mu \neq \lambda$ we may divide by $\mu-\lambda$, and since $\mathbf{w} \neq \mathbf{0}$ (an eigenvector), this forces $d = 0$. Substituting back, $c\mathbf{v} = \mathbf{0}$ with $\mathbf{v} \neq \mathbf{0}$ forces $c = 0$. Both coefficients vanish, so $\mathbf{v}, \mathbf{w}$ are linearly independent. Note the three hypotheses being spent: $\mu \neq \lambda$ (to divide), $\mathbf{w} \neq \mathbf{0}$ and $\mathbf{v} \neq \mathbf{0}$ (to conclude) — the last two are exactly why the definition insists eigenvectors are non-zero. **(b)** Pick one eigenvector per eigenvalue: n distinct eigenvalues give n eigenvectors, and applying (a) pairwise shows the whole set is linearly independent. (The argument extends past pairs: if $\sum c_i\mathbf{v}_i = \mathbf{0}$, apply A and eliminate term by term, or apply $\prod_{j\neq i}(A - \lambda_j I)$ to isolate a single term.) Independent n vectors in an n-dimensional space form a basis, so with $P = [\mathbf{v}_1 \cdots \mathbf{v}_n]$ we get $AP = PD$, P invertible, hence $A = PDP^{-1}$ — diagonalisable. For A of Problem 3 the eigenvalues 5 and 2 are distinct, so it is diagonalisable; Problem 9 exhibits the P and D explicitly. **(c)** No — distinct eigenvalues are sufficient but not necessary. Take $2I$: its only eigenvalue is 2 (algebraic multiplicity 2), yet it is *already diagonal*, so it is trivially diagonalisable, and moreover **every** non-zero vector is an eigenvector. So the genuine obstruction is not a repeated root: it is geometric multiplicity falling short of algebraic multiplicity, which is exactly Problem 7. **(d)** Nothing in the proof uses reality — it only uses subtraction and division in the underlying field. So it is equally valid over $\mathbb{C}$: a real matrix with complex eigenvalues has linearly independent *complex* eigenvectors (Problem 8 is such a case, with eigenvectors involving i).

Problem 7

The shear $S = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$: repeated eigenvalue, missing eigenvector.

(a) $S - \lambda I = \begin{bmatrix} 2-\lambda & 1 \\ 0 & 2-\lambda \end{bmatrix}$ is triangular, so its determinant is the product of the diagonal:

$$\det(S - \lambda I) = (2-\lambda)^2.$$

Single eigenvalue $\lambda = 2$, **algebraic multiplicity 2** (the root appears twice). **(b)** $S - 2I = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$, and $(S - 2I)\mathbf{v} = \mathbf{0}$ says $y = 0$ with x free:

$$\mathbf{v} = t \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \qquad E_2 = \mathrm{Span}\{(1, 0)\}.$$

Only 1 linearly independent eigenvector: the **geometric multiplicity is 1**. Check: $S(1, 0) = (2, 0) = 2(1, 0)$ ✓, while $S(0, 1) = (1, 2)$ is not a multiple of $(0, 1)$. **(c)** Two independent arguments. *Counting.* To write $S = PDP^{-1}$ the columns of P must be $n = 2$ linearly independent eigenvectors (Problem 6(b) reversed). There is only one eigendirection, so no basis of eigenvectors exists. *Contradiction.* Suppose $S = PDP^{-1}$. Diagonal entries of D are the eigenvalues of S (they are visibly the eigenvalues of the diagonal matrix D, and similar matrices share eigenvalues — Problem 10(a)), so every diagonal entry would be 2, giving $D = 2I$ and hence $S = P(2I)P^{-1} = 2I$. But $S \neq 2I$. Contradiction. **(d)** For $2I$ we get $2I - 2I = \mathbf{0}$, whose null space is all of $\mathbb{R}^2$: geometric multiplicity 2, equal to the algebraic multiplicity, so there is a basis of eigenvectors (any basis) and $2I$ is diagonalisable already. What separates them is therefore not the repetition of the root but the **geometric multiplicity** $\dim \mathrm{Nul}(A - \lambda I)$ reaching the algebraic multiplicity; when it falls short — the shear's $1 < 2$ — eigenvalues and eigenvectors alone cannot build a basis. **(e)** In coordinates, $S(x, y) = (2x + y,\ 2y)$. Every point doubles in height and doubles horizontally, then *slides sideways by an amount equal to its new height* — hence "shear": vertical layers slide over each other like a deck of cards. The horizontal axis is the exception, points on it have $y = 0$ and stay horizontal, merely stretched by 2; the line is preserved as a line, so it is the unique eigendirection.

Problem 8

Rotation R, reflection H, rotate-and-scale K.

(a) $\det(R - \lambda I) = \begin{vmatrix} -\lambda & -1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 + 1$, so $\lambda^2 = -1$ and $\lambda = \pm i$. No real eigenvector exists for a purely geometric reason: an eigenvector would have to satisfy $R\mathbf{v} = \lambda\mathbf{v}$, meaning $\mathbf{v}$ stays on its own line through the origin. But R rotates every non-zero vector by exactly 90°, which moves every line onto a different line. Confirm: $R(x, y) = (-y, x)$ is parallel to (x, y) only if $(-y, x) = c(x, y)$, giving $-y = cx$ and $x = cy$, hence $x = c(-cx) = -c^2 x$ and $x(1 + c^2) = 0$, so $x = 0$ and then $y = 0$. Eigenvalues $\pm i$; no real eigendirection — the rotation has no invariant line. **(b)** $\det(H - \lambda I) = \begin{vmatrix} -\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = \lambda^2 - 1 = (\lambda - 1)(\lambda + 1)$, so $\lambda = \pm 1$ (real!). Then $H(1, 1) = (1, 1) \to \lambda = 1$, and $H(1, -1) = (-1, 1) = -(1, -1) \to \lambda = -1$. Both matrices have trace 0 — the difference is the determinant: $\det R = 0 \cdot 0 - (-1)(1) = 1$ while $\det H = 0 \cdot 0 - 1 \cdot 1 = -1$. For a 2×2 matrix the discriminant of the characteristic polynomial is $(\mathrm{tr}\, A)^2 - 4 \det A$: here $0 - 4(1) = -4 < 0$ for R (complex pair) versus $0 - 4(-1) = 4 > 0$ for H (two real roots). Geometrically, a reflection *does* leave two lines in place (the mirror line fixed, the perpendicular line reversed), and those lines are precisely its eigenvectors. **(c)** $\det(K - \lambda I) = (1 - \lambda)^2 + 1 = \lambda^2 - 2\lambda + 2$, whose roots are $\lambda = 1 \pm i$. In polar form $1 \pm i = \sqrt{2}\, e^{\pm i\pi/4}$, so $r = \sqrt{2}$ and $\theta = \pi/4$: predict that K *scales every length by* $\sqrt{2}$ *and rotates by* 45° counter-clockwise. Test on $(1, 0)$:

$$K \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad \big|(1, 1)\big| = \sqrt{2}\ \checkmark, \qquad \text{angle from } (1, 0) = 45^\circ\ \checkmark.$$

The prediction holds. **(d)** Dictionary: writing the pair as $a \pm bi = re^{\pm i\theta}$ with $r = \sqrt{a^2 + b^2}$ and $\theta = \arctan(b/a)$, the matrix acts as **scale by** r**, rotate by** θ — the single sentence behind

"complex eigenvalues mean rotation is involved". A large and important class never produces this: **real symmetric matrices** ($A^T = A$) always have real eigenvalues (the spectral theorem), which is why B in Problem 2, H above, and any covariance matrix you meet later have only real spectra.

Problem 9

Diagonalising $A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ and computing A^k by hand.

(a) Work column by column. Column j of AP is A applied to column j of P, i.e. $A(1,1)$ and $A(1,-2)$:

$$AP = \begin{bmatrix} 5 & 2 \\ 5 & -4 \end{bmatrix}; \qquad PD = \begin{bmatrix} 1 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} 5 & 0 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 5 \cdot 1 & 2 \cdot 1 \\ 5 \cdot 1 & 2(-2) \end{bmatrix} = \begin{bmatrix} 5 & 2 \\ 5 & -4 \end{bmatrix}.$$

They agree. And this is no coincidence: column j of PD is d_j times column j of P, so the j-th column of $AP = PD$ reads exactly "$A\mathbf{p}_j = d_j\mathbf{p}_j$" — the statement that each column of P is an eigenvector with eigenvalue d_j. Recognising a diagonalisation therefore costs only n eigenvector checks; you never have to invert anything. **(b)** $\det P = 1(-2) - 1 \cdot 1 = -3$, so $P^{-1} = \frac{1}{-3}\begin{bmatrix} -2 & -1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{bmatrix}$. Then, reusing PD from (a),

$$PDP^{-1} = \begin{bmatrix} 5 & 2 \\ 5 & -4 \end{bmatrix} \begin{bmatrix} 2/3 & 1/3 \\ 1/3 & -1/3 \end{bmatrix} = \begin{bmatrix} \frac{10}{3} + \frac{2}{3} & \frac{5}{3} - \frac{2}{3} \\ \frac{10}{3} - \frac{4}{3} & \frac{5}{3} + \frac{4}{3} \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} = A \ \checkmark.$$

(c) Decompose the vector in the eigenbasis: $(2,-1) = (1,1) + (1,-2)$ (check: $1+1=2$, $1-2=-1$). Since A multiplies the first piece by 5 and the second by 2 at every application,

$$A^k \begin{bmatrix} 2 \\ -1 \end{bmatrix} = 5^k \begin{bmatrix} 1 \\ 1 \end{bmatrix} + 2^k \begin{bmatrix} 1 \\ -2 \end{bmatrix} = \begin{bmatrix} 5^k + 2^k \\ 5^k - 2^{k+1} \end{bmatrix}.$$

Check $k=1$: $(5+2,\ 5-4) = (7,1)$, and indeed $A(2,-1) = (8-1,\ 4-3) = (7,1)$ $\checkmark$. Check $k=2$: $(25+4,\ 25-8) = (29,17)$, while $A^2 = \begin{bmatrix} 18 & 7 \\ 14 & 11 \end{bmatrix}$ and $A^2(2,-1) = (36-7,\ 28-11) = (29,17)$ $\checkmark$. **(d)** Divide by 5^k: $\dfrac{A^k(2,-1)}{5^k} = (1,1) + \left(\dfrac{2}{5}\right)^k (1,-2) \longrightarrow (1,1)$ as $k \to \infty$, since $(2/5)^k \to 0$. In general, if $\mathbf{x} = \sum c_i \mathbf{v}_i$ with $A\mathbf{v}_i = \lambda_i \mathbf{v}_i$ and $|\lambda_1|$ the largest modulus, then $A^k\mathbf{x} = \lambda_1^k (c_1\mathbf{v}_1 + \sum_{i>1} c_i (\lambda_i/\lambda_1)^k \mathbf{v}_i)$, and every ratio $(\lambda_i/\lambda_1)^k$ decays. So the iterated process is dominated by the largest eigenvalue, and the surviving direction is $(1,1)$ — the $\lambda = 5$ eigendirection. This single observation is the engine behind Markov chains and steady states: repeated application washes out everything except the dominant eigenmode.

Problem 10

Similarity: what survives $B = P^{-1}AP$.

(a) Insert $I = PP^{-1}$ inside the λ term:

$$\det(B - \lambda I) = \det(P^{-1}AP - \lambda P^{-1}P) = \det\big(P^{-1}(A - \lambda I)P\big) = \det(P^{-1}) \det(A - \lambda I) \det(P) = \det(A - \lambda I),$$

since $\det(P^{-1})\det(P) = \det(I) = 1$. Identical characteristic polynomials mean identical eigenvalues with identical algebraic multiplicities. **(b)** If $A\mathbf{v} = \lambda\mathbf{v}$ then

$$B(P^{-1}\mathbf{v}) = P^{-1}AP\,P^{-1}\mathbf{v} = P^{-1}A\mathbf{v} = P^{-1}(\lambda\mathbf{v}) = \lambda(P^{-1}\mathbf{v}),$$

so the eigenvector of the similar matrix B is $P^{-1}\mathbf{v}$ — the same vector expressed in P's coordinates, not the same list of numbers. Concretely, with A and P from Problem 9 the similar matrix is $B = P^{-1}AP = \begin{bmatrix}5 & 0\\ 0 & 2\end{bmatrix}$ and $P^{-1} = \begin{bmatrix}2/3 & 1/3\\ 1/3 & -1/3\end{bmatrix}$:

$$P^{-1}\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}1\\0\end{bmatrix} \checkmark \text{ (eigenvector of } B \text{ for } \lambda = 5), \qquad P^{-1}\begin{bmatrix}1\\-2\end{bmatrix} = \begin{bmatrix}0\\1\end{bmatrix} \checkmark \text{ (for } \lambda = 2).$$

The eigenvectors became the coordinate axes — the cleanest possible illustration that eigenvectors are basis-dependent while eigenvalues are not. **(c)** Take $S = \begin{bmatrix}2 & 1\\ 0 & 2\end{bmatrix}$ (Problem 7) and $2I$. Both have characteristic polynomial $(\lambda - 2)^2$: identical spectrum, same multiplicities. Yet they are not similar. For if $S = P(2I)P^{-1}$ then $S = 2I$, false. Equivalently, geometric multiplicity is invariant under similarity (multiplication by P^{-1} is an isomorphism, so it preserves null-space dimensions), and it is 1 for S versus 2 for $2I$. Same eigenvalues $\not\Rightarrow$ similar. The defect (algebraic minus geometric multiplicity) is itself a similarity invariant. **(d)** Invariant under similarity: characteristic polynomial, eigenvalues with algebraic multiplicities, trace, determinant, rank, nullity, invertibility, and geometric multiplicities. Not invariant: the eigenvectors themselves, the entries, symmetry, orthogonality. Conceptually: a matrix is a *description* of one linear map in some coordinate system, and change of basis replaces A by $P^{-1}AP$. Anything that belongs to the map itself — the numbers λ for which $(T - \lambda I)$ fails to be invertible — cannot depend on coordinates. Anything expressed in coordinates, above all the vectors themselves, must change: they are translated into the new basis, exactly as $P^{-1}\mathbf{v}$ in part (b).

Problem 11

True or false, each with a reason.

(i) False. $R = \begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}$ has $\lambda^2 + 1 = 0$, so its eigenvalues are $\pm i$ and it has no real eigenvalue at all. What *is* true: over $\mathbb{C}$ every square matrix has at least one eigenvalue, because $\det(A - \lambda I)$ is a degree-n polynomial and the fundamental theorem of algebra gives it a root. **(ii) True.** Exactly Problem 5(a): $A\mathbf{v} = 0\mathbf{v} = \mathbf{0}$ with $\mathbf{v} \neq \mathbf{0}$ means a non-trivial null space, i.e. A singular, and conversely a singular A has some non-zero $\mathbf{v}$ with $A\mathbf{v} = \mathbf{0}$. **(iii) True.** $(A + 3I)\mathbf{v} = A\mathbf{v} + 3\mathbf{v} = \lambda\mathbf{v} + 3\mathbf{v} = (\lambda + 3)\mathbf{v}$, and $\mathbf{v} \neq \mathbf{0}$. Shifting by a multiple of the identity shifts the spectrum without touching the eigenvectors. **(iv) False.** With $A = \begin{bmatrix}4 & 1\\ 2 & 3\end{bmatrix}$ and $\mathbf{v} = (1, 1)$: $A^T(1, 1) = (4 \cdot 1 + 2 \cdot 1,\ 1 \cdot 1 + 3 \cdot 1) = (6, 4)$, which is not a multiple of $(1, 1)$. What A and A^T *do* share is the eigenvalues: $\det A^T = \det A$ and $\operatorname{tr} A^T = \operatorname{tr} A$, and indeed A^T has the same characteristic polynomial, with $\lambda = 5$ belonging to $(2, 1)$ — check $A^T(2, 1) = (10, 5) = 5(2, 1)$ — and $\lambda = 2$ belonging to $(-1, 1)$ — check $A^T(-1, 1) = (-2, 2) = 2(-1, 1)$. **(v) False.** Same counterexample as Problem 4(c): $(1, 1) + (1, -2) = (2, -1)$ but $A(2, -1) = (7, 1)$, which is not parallel to $(2, -1)$. The corrected rule: sums are safe only *within one eigenspace*. **(vi) True.** Problem 6(a) — provided the eigenvalues are distinct, which is the whole content of that proof. **(vii) False.** Problem 10(c):

the shear $\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$ and $2I$ share $(\lambda - 2)^2$ but are not similar, because no invertible P can repair a missing eigenvector.

Spotted an error, or a solution that assumes something it should not? Open an issue on LA-hw and quote the problem and part — corrections are welcome.